SAMRATASHOKTECHNOLOGICALINSTITUTE
(Engineering College), VIDISHA M.P.
(An Autonomous Institute Affiliated to RGPV Bhopal)
Computer Science and Engineering

Semester/Year			Program		B.Tech.
Subject Category	DLC	Subject Code:	**CS26102P**	Subject Name:	**Advance Programming**

Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		Total Marks				
End Sem	Mid-Sem	Assignment / Quiz	End Sem	Lab- Work		L	T	P	
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Course Objective:

- The objective of this course is to provide students with a strong foundation in procedural and object-oriented programming using C and C++.
- To develop a strong understanding of advanced C programming concepts including pointers, dynamic memory allocation, and file handling.
- To provide knowledge of C++ programming fundamentals such as data types, variables, operators, expressions, and control structures.
- The course aims to develop problem-solving skills through advanced concepts of pointers, memory management, file handling, classes and objects, inheritance, polymorphism, templates, and dynamic memory allocation.
- To enhance problem-solving and programming skills for developing efficient and structured software applications using C and C++.
- Students will be able to design, implement, and manage efficient software solutions using modern programming techniques.

Course Outcomes:

CO1: Apply advanced C programming concepts including pointer arithmetic, multidimensional arrays, function pointers, dynamic memory allocation, and file handling for solving programming problems.

CO2: Demonstrate proficiency in C++ programming fundamentals including data types, variables, operators, control structures, standard libraries, and random number generation.

CO3: Design and implement object-oriented programs using classes, objects, constructors, function overloading, and appropriate access specifiers.

CO4: Develop software applications utilizing inheritance, abstract classes, virtual functions, pointers to objects, and dynamic memory management to achieve code reusability and polymorphism.

CO5: Create generic and reusable solutions using class templates, function templates, and member function templates while applying object-oriented design principles effectively.

UNITS	Descriptions	Hrs.	CO's
I	**Pointer and Memory Management(C):** Deep dive into pointer arithmetic, multi-dimensional arrays, function pointers, and dynamic memory allocation (malloc, calloc, realloc, free).	4	CO1
II	**Introduction to OOP, Classes, and Objects (C++):** Basic Concepts of OOP - Benefits of OOP. Class and object fundamentals and various visibility modes in class, Data Types, Variables and Constants, Operators, Arithmetic and Logical Expressions, Assignment Statements and Type Casting.	4	CO2
III	**Control structures-** Iteration statements, Jump Statements and Selective statements. Common C++ Header files, Object as function arguments.	4	CO3
IV	**Constructor**, Default Constructor, Overloaded Constructor, Copy Constructor and Constructor with default arguments. Destructors. Concept of Overloading.	4	CO4
V	**Inheritance and Functions (C++):** Introduction to inheritance and its types. Friend Functions, Abstract Classes. Pointers to Objects – THIS Pointer - Virtual Functions	4	CO5
Guest Lectures (if any)		--	
Total Hours			20

List of Experiments

1. Write a program to modify a character pointer's address using a double pointer (char **).(CO1)
2. Implement a program to dynamically allocate a 1D array using malloc() and calculate the average of elements.(CO1)
3. Implement a text-based calculator menu using an array of function pointers.(CO1)
4. Write a program to create a text file and write student details into it.(CO2)
5. Write a program to store and retrieve student records using structures and files.(CO2)
6. Write a program to create a file and check whether it exists before performing operations.(CO2)
7. Write a program to store and read structure data using binary files (fwrite() and fread()).(CO2)
8. Write a program to Find the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of two numbers.(CO3)
9. Write a Program to print a hollow and diamond pyramid pattern.(CO3)
10. Write a program to reverse a number and check whether it is a palindrome.(CO3)
11. Write a program to find the day of the week using a menu-driven switch statement.(CO3)
12. Write a program to create a class Student with data members Roll No, Name, and Marks. Accept and display the details. (CO4)
13. Write a program to create a class Rectangle with multiple constructors: (CO4)
    - No argument
    - One argument
    - Two arguments
14. Write a program to Overload a function area() to calculate: (CO4)
    - Area of a square
    - Area of a rectangle
    - Area of a circle
15. Demonstrate ambiguity caused by improper use of function overloading and default arguments.(CO4)
16. Create a base class Employee and derive Teacher, Clerk, and Accountant.(CO5)
17. Design a program demonstrating hybrid inheritance for a university management system.(CO5)
18. Create an abstract class Shape with a pure virtual function area() and derive Circle and Rectangle.(CO5)
19. Implement an online payment system using abstract classes and pure virtual functions.(CO5)
20. Create a class template for storing and displaying data of different data types.(CO5)

TextBooks-

1. "Let us 'C'" by YashvantKanitkar ,BPB Publications.
2. "Programming in ANSI C", 7thEdition, 2017, by E. Balagurusamy, McGraw Hill Education India Private Limited.
3. E Balagurusamy, Object Oriented Programming with C++, McGraw Hill Education.

REFERENCESBooks-:

1. Ravichandran, Programming With C++, Tata McGraw Hill.
2. Dromey G, How to solve it by Computer, Prentice Halll – 1978.
3. "Programming with C" by Gottfied.B.S ,Schaum"s outline series, TMH 2001.

Modes of Evaluation and Rubric

The evaluation modes consist of performance in Quiz/ Assignments, term work, and end-semester practical examinations.

List/Links of e-learning resource

- https://onlinecourses.nptel.ac.in
- https://www.classcentral.com(swayam)

Recommendation by Board of studies on	June-2026
Approval by Academic council on	June-2026
Compiled and designed by	CS& AI
Subject handled by department	CS&AI